Supplement: Execution State and Cost Boundaries of Reference-Preserving Speculative Decoding

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1. Reading Guide and Historical Evidence

This supplement retains the late contracts and adds historical studies used in the journal argument. Panels are not pooled; new calculations use saved exports without inference. The package binds historical extracts to repository blobs and late evidence to its inherited manifest. Bindings support content consistency and public arithmetic, not independent authentication of physical execution.

Table 1: Earlier studies restored to the argument. Native, prediction, and retrospective record counts have distinct meanings.

Study	Units and outcomes	Interpretation
B570 intervention	24 exposed tasks; 72 arrays per condition; reference 60/72 to 72/72	Four repairs, 20 controls; compound handoff
RTX intervention	19 HumanEval and 24 MBPP tasks; three repetitions per condition	Prediction closes; one repair and one regression
RTX target-only bridge	19 exposed tasks; 171 fresh runs	$U_2 = N_1$ on 57 arrays/boundaries; no speed claim
Offline selection	RTX: 114/114; B570: 72/144 record equivalents	No supported extra capability; not latency
RTX global serialization	19 exposed tasks; 171 fresh timing observations	Qualifies 57/57, but 0.677 times target-only speed

Early prediction/native comparisons use complete capped arrays; equal EOS/limit reason alone is not strict boundary equality. The bridge additionally checks committed length and terminal ID. HumanEval/15 changes

length 25 to 28; /25 stays at 75 IDs while changing at index 4; /18 stays at 69 divergent IDs. Sixty uncommitted target samples remain outside committed-output equality. Historical U2 threads differ from the fresh bridge; matching output is observational, not thread causality.

Selection counts are acquisition equivalents after common reference/baseline work. Proxy screening has no demonstrated timing advantage. Unknown counterfactuals are not filled from another study; MBPP U2 remains unavailable.

The serialized RTX cost arm has a synchronized post-prefill-to-slot-release timer, condition medians over three repetitions, and a task geometric mean. Its 0.677 speed and [0.662,0.692] descriptive interval concern instrumented execution including logging. It cannot be directly compared with the later generation timer as an improvement trend.

2. Study Identities and Independence

This supplement separates discovery, qualification, timing, and post-hoc reconstruction. Repeated observations are never counted as new inputs. Public derived-table validation is distinct from original-host raw verification. The source manifest outside the author-review PDF binds exact project commits and files; those identifiers are omitted here to avoid revealing authorship.

The RTX pilot includes four already exposed HumanEval tasks with qualified speedup under some native configurations, then expanded screening failures. In its final eight-task check, small-model $k = 2$ qualifies 18/24 repeated observations. A shorter output in a failed mixed strategy is not counted as acceleration. The pilot's FA-on/UB512 reference can differ from the later fixed FA-off scalar reference. This history motivates the constrained-path experiment but cannot substitute for its qualification.

The initial transfer plan sought unused HumanEval and MBPP inputs. Accessible exposure accounting rules out an unused HumanEval panel at that stage. Before new outputs, the protocol instead freezes 24 MBPP inputs. Their status is unused in the accessible experiment history, not a claim about training contamination. Once measured, the same inputs are exposed for all later post-hoc analyses.

3. RTX Expansion: Statistics and Timer Boundaries

For each task and condition, take four absolute generation or request times and compute their median. Compute the baseline/candidate ratio from those

Table 2: Late study sequence. Panels overlap unless explicitly identified as project-unused inputs. None is a new model-family replication.

Study	Units	Evidence role and limit
Execution minimization plan	1005 inherited records	CPU plan/audit; no new native causal experiment.
RTX efficiency pilot	300 formal requests	Screening under its own references; 48 excluded warmups; not pooled with later costs.
RTX natural state crosses	4 probes; 24 donors, 48 forks, 24 shams	Selected-boundary causal interventions; auxiliary forwards not latency measurements.
RTX scope factorial	16 masks, 4 probes, 2 repeats	128 requests; passing masks 3/7/11/15 within the fixed family.
RTX scope cost	19 exposed tasks, 3 conditions, 6 repeats	342 formal requests; 18 excluded warmups; capture off.
RTX input expansion	24 eligible MBPP tasks, 4 conditions, 4 repeats	384 formal requests; 16 excluded warmups; 96 qualifications per condition.
RTX fixed-UB width	28 exposed qualification tasks, 2 repeats; separate cost sessions	Qualification selects $k = 7$; width cost is measured independently of diagnostic capture.
B570 finite mechanism	6 repeated crosses over 3 tasks, 12 shams	State/whole-recipe observations; MBPP/72 current columns duplicate.
B570 expanded cost	15 project-unused tasks, 4 conditions, 2 repeats	E119/120; R/M/C each 30 complete; old-N1 missing one payload.
RTX/B570 CPU closeouts	Existing saved observations only	No new inference, hardware replication or untouched test panel.

medians, then its unweighted geometric mean across 24 tasks. Do not take a ratio of cross-task median times, pool repeated requests as independent inputs, or switch numerator direction to make a speedup.

The primary descriptive bootstrap uses NumPy Generator PCG64, seed 20261008, 10,000 task resamples of 24 indices, with one common index matrix across comparisons. The interval uses linear/type-7 quantiles. Historical intervals from Python `random.Random`, seed 20261005, 4,000 draws and sorted indices 99/3899 are reproduced separately. Differences between the two conventions are not new measurements.

Table 3: All RTX expansion pairwise summaries. Speed is baseline/candidate. Each row uses 24 task medians; candidate conditions are output-qualified.

Baseline / candidate	Generation	Resident request
Ordinary / mask 3 speculation	1.170124	1.741569
Matched target / mask 3 speculation	1.169238	1.035071
Ordinary / matched target	1.000758	1.682560
Ordinary / mask 15 speculation	1.152348	1.690202
Matched target / mask 15 speculation	1.151475	1.004542
Mask 15 / mask 3 speculation	1.015426	1.030391

The scalar reference uses mask 0/UB1 without draft. The matched condition uses mask 3/UB4 without draft. The two speculative conditions use mask 3 or mask 15 with $k = 3$/UB4. All conditions use the same frozen private server and retained CUDA libraries. Formal request capture is disabled. Each round uses four separate resident processes and one warmup per process; condition order and task order rotate. Actual recorded cache reuse is zero. The response's `tokens_cached` field describes the final slot prompt length in this source; it is not the cache-hit count.

Generation time is the retained server `predicted_ms` field between its prompt/first-sample boundary and its last generation update. It charges drafting/verification within that interval, excludes prompt work, and can exclude later final-token bookkeeping. Resident request time is the client wall time around complete non-streaming HTTP response receipt and parsing. Process startup, model load, excluded warmups, boundary telemetry and server shutdown lie outside it. Post-response asynchronous cleanup is not certified to be inside it.

No cost fractions are inferred from `prompt_ms + predicted_ms` or request

minus generation. The state-transplant helper is conditional in the source, but its activating environment variable is absent in this expansion. A dormant branch cannot be charged as an observed transfer. The ordinary-to-matched difference changes prefill microbatch/attention and continuation capacity jointly. Only separate component instrumentation could establish an attention, transport or transfer decomposition.

4. State Crosses and Scope Interpretation

Natural donors are reached with native proposals rather than a recorded proposal stream supplied to the candidate. Equal committed prefix and valid cache ownership are checked before crossing. Own-path shams test manipulation fidelity. Selected-boundary studies are deliberately informative about known sensitivity; they cannot estimate how common the effect is among arbitrary requests.

In the first RTX cross, /15 changes with the current call under both donors, /25 changes only in one donor/current combination, and /18 and /31 do not change at their selected cuts. Serialized donor states differ, while prompt logical K/V agree. Generated-prefix differences are not a unique attention attribution: numerical inputs to later computation may already differ. Continuous score changes and discrete argmax changes are separate observables.

Scope bits target prefill/continuation and draft prefill/continuation encode a finite candidate family. Passing masks 3/7/11/15 show that draft constraints are unnecessary for this panel. They do not show that every implementation can leave draft execution arbitrary without satisfying the usual verification/state/commit conditions. Mask 3's minimality is an inclusion relation in the tested switch family, not a theorem over all possible numerical transformations.

The expanded MBPP/229 wide-geometry cross flips the EOG decision with natural donor state under both tested layouts. It supplies a state-sensitive failure boundary for the fixed repair. The two layouts also change continuous values; absence of a separate argmax flip does not mean they are numerically equivalent. Two cap contexts, four shams and auxiliary probes are retained as repeated observations of one task.

For B570, the recipes include both microbatch and output-reservation differences. The /145 and /42 matrices support whole-recipe interaction, not isolated microbatch causality. /72 has identical current columns and is a

state control. Six repeated cases are three tasks. No full counterfactual suffix from each crossed state is claimed. Full repaired requests have their own qualification matrix.

5. B570 Endpoint, Missingness, and Work

The CPU closeout recomputes qualification, timer bindings, work and causal summaries without inference. Its cap/EOG contract preserves endpoint agreement while retaining the distinction from raw-array equality. On the 15-input formal panel, ordinary, matched and candidate each supply 30 complete outputs. Old N1 lacks one payload, so the full matrix stays 119/120. Aggregate old-N1 comparisons use 14 complete task pairs rather than filling the missing request.

The candidate drafts 888 and accepts 702 tokens in the complete formal candidate records. Target continuation caller tokens are 1332, versus 1120 for matched target; draft continuation caller tokens are 1362. Endpoint tokens total 1150 while raw tokens total 1176, with 26 extra suffix tokens. These counts describe different runtime quantities. Caller tokens are not graph counts or kernel launches. Neither token count nor ratio differences identify an independently timed component share.

Reference/candidate speed is 0.282588 generation and 0.201290 request; matched/candidate is 0.285761 and 0.20570 approximately. Old-N1/candidate is 3.99309 and 2.63394 over 14 complete task-median pairs. The old-N1 request comparison reverses direction in one individual round at MBPP/245. It remains incorrect to report either universal old-path improvement or ordinary-reference gain.

6. Historical Baselines and Abandoned Directions

Earlier RTX target-continuation interventions were predicted on 43 tasks but repaired one task while introducing another deviation, leaving aggregate reference agreement unchanged. The ordinary draft-free U2 bridge reproduces all 19 measured intervention outputs, with reference agreement 17/19. An offline intervention-selection study finds no measured advantage over tested simple alternatives. These negative findings concern the earlier predictor/selector candidates; they do not invalidate the later native attention-layout instance.

Early RTX heterogeneous and native costs, and a 14B configuration with output failures, remain source-specific historical observations. Unqualified speed cannot establish losslessness. No historical timing is paired with a later output trace to create a synthetic qualified speedup. Detailed older experiment exports remain in the versioned historical manuscript archive.

The M4 prospective 60-task study is an actual post-prefill campaign, distinct from earlier proxy timings. Always qualifies 180/180 and reaches 1.346894; Guard qualifies 180/180 and reaches 1.311386. Guard/Always latency is 1.027076 with descriptive 95% interval[1.012956,1.043559]. A per-task whole-request oracle choosing only between target-only and Always leaves approximately 0.0691967% further improvement relative to Always. This is not an intra-request controller ceiling.

On 358H, measured CPU/NPU drafting yields 0.733915/0.351663 with 90/90 reference agreement each. Available conditional overhead-removal ceilings remain below parity. The historical NPU full-prefix replay explains an implementation cost, while incomplete tracing prevents a universal component model. M4 and 358H support cost conclusions under their own contracts; they do not inherit the late natural-state evidence.

7. Reproducibility and Remaining Gaps

The JSA v0.13.0 reconstruction retains all 26 inherited evidence files and three existing figures byte-for-byte. Its build checks public RTX qualification-hash joins, median-before-ratio costs, descriptive intervals, and two-baseline identities. Additional CPU checks reproduce B570 paid costs and M4/RTX workload summaries. The source package binds historical extracts to the latest repository blobs. No runtime or GPU is called and no figures are regenerated.

The earlier integration's 4,320-entry audit and editorial figure-source checks remain historical verification records. Current calculations reproduce saved public arithmetic, not private raw-output authentication. Independent archive replay, new hardware measurements, nearest-method implementation comparisons, cross-family mechanisms, stochastic sampling, and concurrency remain outside this baseline.

8. Additional Workload and Cost Interpretation

Fixed-UB width costs reuse 24 MBPP and four HumanEval qualification inputs in four-repetition capture-off sessions: 448 requests and 16 excluded

warmups. Width is selected by qualification, not speed. The build recomputes subgroup and pooled task-ratio geometric means before generating the manuscript table. Four exposed HumanEval tasks are not a generalization panel.

M4's HumanEval and Spec-Bench partitions each contain 30 prospective inputs. Fixed and guarded speedups use saved absolute task medians. The whole-request oracle chooses only target-only or Always; it does not characterize intra-request adaptation, different implementations, or production optimality.

For 358H, pooled-total and prompt median-of-paired-ratios ceilings answer different aggregation questions. With reference cost, target work, and progress fixed, deleting other nonnegative work leaves speed bounded by target-work-only ratios. The ceilings are conditional budgets, not achieved savings or architectural impossibility.

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